

Reviewer Pack — Minimal Quadratic Discriminant Correlation with Frege Proof ...

Entry 3077415568d5 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Quadratic Discriminant Correlation with Frege Proof Length

Entry ID: 3077415568d5 **Verdict:** INCONCLUSIVE **Recorded:** 2026-06-01 20:17:36 UTC

1. Conjecture

Field A (mathematical branch): Quadratic Number Theory **Field B** (complexity object): Frege Proof Complexity

Statement:

For every Boolean formula φ , the minimal discriminant δ of its polynomial representation in the quadratic number field is linearly correlated with its Frege proof length $f(\varphi)$, such that $\log(\delta) = \Theta(f(\varphi))$. Equivalently, for all instances φ with a fixed discriminant range, there exists a constant $c > 0$ such that $\log(\delta) = O(c * f(\varphi))$ and $\Omega(c * f(\varphi))$.

Rationale (proposer's reasoning):

Quadratic Number Theory provides an algebraic structure to represent Boolean functions, while Frege proofs are known to be an efficient method for proving tautologies. A correlation between the discriminant of a polynomial representation and the proof length could expose a deeper connection between algebraic structures and computational complexity.

Taxonomy category: Quadratic Number Theory × Frege Proof Complexity (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): e06d6ed52421bfe8

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

For a given CNF formula φ , the minimal discriminant δ and Frege proof length $f(\varphi)$ are considered supported if the correlation coefficient r^2 from a simple linear regression is ≥ 0.9 for all seeds and the p-value is ≤ 0.01 .

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	1.00	SAFE	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=1, elapsed=0.1s

5.1 Generated Python source

```
import random
import math
from fractions import Fraction

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_cnf(n):
        clauses = []
        for _ in range(n):
            clause = [random.choice([-1, 1]) * (i + 1) for i in range(n)]
            clauses.append(clause)
        return clauses

    def cnf_to_poly(cnf):
        n = len(cnf[0])
        poly = [[Fraction(0)] * n for _ in range(n)]
        for clause in cnf:
            for lit in clause:
                var = abs(lit) - 1
                if lit > 0:
                    poly[var][var] += Fraction(1)
                else:
                    poly[0][var] -= Fraction(1)
        return poly

    def determinant(matrix):
        n = len(matrix)
        det = Fraction(0)
        for i in range(n):
            submatrix = [[matrix[j][k] for k in range(j + 1, n)] for j in range(i + 1, n)]
            sign = (-1) ** i
            if n == 2:
                det += sign * (matrix[0][0] * matrix[1][1] - matrix[0][1] * matrix[1][0])
            else:
                det += sign * matrix[i][0] * determinant(submatrix)
        return det
```

```

def discriminant(poly):
    n = len(poly)
    if n == 2:
        a, b, c = poly[0][0], poly[0][1], poly[1][1]
        return b**2 - 4*a*c
    else:
        return determinant([[poly[i][j] for j in range(i + 1, n)] for i in range(n)])

def frege_proof_length(cnf):
    # Placeholder function; actual implementation depends on the Frege proof system
    return len(cnf) * 2

results = []
for n in [5, 10, 15, 20, 30, 40]:
    for _ in range(5): # Ensure at least 30 instances per seed
        cnf = generate_cnf(n)
        poly = cnf_to_poly(cnf)
        delta = discriminant(poly)
        f_phi = frege_proof_length(cnf)
        if delta <= 0:
            continue
        results.append((math.log(delta), f_phi))

if not results:
    return {
        "metric_name": "log(discriminant)",
        "metric_value": None,
        "instances_tested": 0,
        "n_max": 0,
        "conjecture_holds": False,
        "counterexample": "mapping_undefined"
    }

log_delta, f_phi = zip(*results)
n = len(log_delta)
mean_log_delta = sum(log_delta) / n
mean_f_phi = sum(f_phi) / n
variance_log_delta = sum((x - mean_log_delta) ** 2 for x in log_delta) / n
variance_f_phi = sum((x - mean_f_phi) ** 2 for x in f_phi) / n

correlation_coefficient = (sum((log_delta[i] - mean_log_delta) * (f_phi[i] - mean_f_phi) for i in range(n)) /
                           math.sqrt(variance_log_delta * variance_f_phi))

return {
    "metric_name": "log(discriminant)",
    "metric_value": correlation_coefficient,
    "instances_tested": n,
    "n_max": max([len(cnf) for cnf in results]),
    "conjecture_holds": correlation_coefficient >= 0.9 and p_value <= 0.01,
    "counterexample": ""
}

if __name__ == "__main__":
    import sys
    seeds = [int(x) for x in sys.argv[1:]] or [2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 67, 71, 73, 79, 83, 89, 97]

```

```

results = []

for seed in seeds:
    trial_result = run_trial(seed)
    print(f"TRIAL: {trial_result}")
    results.append(trial_result)

mean_metric_value = sum(result["metric_value"] for result in results if result["metric_value"]
std_metric_value = math.sqrt(sum((result["metric_value"] - mean_metric_value) ** 2 for result
support_fraction = sum(1 for result in results if result["conjecture_holds"]) / len(results)

if all(result["conjecture_holds"] for result in results):
    print(f"RESULT: SUPPORTED mean={mean_metric_value} std={std_metric_value} support_fraction")
elif support_fraction >= 0.8:
    print(f"RESULT: SUPPORTED mean={mean_metric_value} std={std_metric_value} support_fraction")
else:
    for result in results:
        if not result["conjecture_holds"]:
            print(f"RESULT: FALSIFIED counterexample=\"not supported\" first_failing_seed={res
            break

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```

--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_879ad864.py", line 110, in <module>
    trial_result = run_trial(seed)
                    ~~~~~^~~~~~
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_879ad864.py", line 69, in run_trial
    delta = discriminant(poly)
            ~~~~~^~~~~~
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_879ad864.py", line 58, in discriminant
    return determinant([[poly[i][j] for j in range(i + 1, n)] for i in range(n)])
                       ~~~~~^~~~~~
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_879ad864.py", line 44, in determinant
    submatrix = [[matrix[j][k] for k in range(j + 1, n)] for j in range(i + 1, n)]
                  ~~~~~^~~~~~
IndexError: list index out of range

```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test code crashed before producing data, which means it did not complete the computation to determine the correlation coefficient and p-value. | next: Re-run the test with proper error handling to ensure that it completes execution and produces the necessary data for analysis.

11. Audit log (LLM calls)

Total LLM calls: 10

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	13339
2	propose	ollama_remote	glm4:latest	0	0	22886
3	preregistration	ollama_remote	glm4:latest	0	0	9259
4	novelty	ollama_remote	glm4:latest	0	0	9701
5	novelty	ollama_remote	glm4:latest	0	0	13196
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	27312
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	17149
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	18687
9	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	19611
10	judge	ollama_remote	glm4:latest	0	0	11128

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 162267 ms total latency. Provider mix: {'ollama_remote': 10}

(full prompt+response transcripts available in *research/audit/3077415568d5.jsonl*)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in *research/audit/3077415568d5.jsonl* for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: *research/pvsnp_sandbox/test_*.py* (per-cycle)

Replay tarball: *research/replay/3077415568d5.tar.gz* (if generated)